PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE NAME	

CT GROUP	1	4				INDEX NUMBER						

CHEMISTRY 9647/01

Paper 1 Multiple Choice **25 September 2015**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, CT group and index number in the spaces at the top of this page and on the Answer Sheet in the spaces provided.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions in this paper. Answer **all** questions.

For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 A solid hydrocarbon was completely combusted in a closed vessel at 120 ^{o}C. The residual gas had a volume of 64 cm^3, which decreased by 24 cm^3 after bubbling through a dehydrating agent. After this, 40% of the final gas volume consisted of oxygen.

What is the empirical formula of the hydrocarbon?

A CH

B CH_2

C CH_3

D C_2H_3

2 When 20 cm^3 of 0.200 mol dm^{-3} KI(aq) was added to 5 cm^3 of 0.200 mol dm^{-3} $CuSO_4$(aq) in a beaker, a white precipitate in brown solution was obtained according to the equation shown below.

$$2Cu^{2+} + 4I^{-} \rightarrow 2CuI + I_2$$

What is the volume of 0.040 mol dm^{-3} $Na_2S_2O_3$(aq) that should be added to the mixture to completely decolourise the brown solution?

A 12.5 cm^3

B 15.0 cm^3

C 16.7 cm^3

D 25.0 cm^3

3 Which of the following statement explains why the boiling point of water (100 °C) is higher than that of ammonia (–33 °C)?

A Ammonia has intramolecular hydrogen bonds, which water does not have.

B The M_r of water is greater than that in ammonia, so van der Waals' forces are stronger in water.

C There are, on average, more hydrogen bonds between water molecules than there are between ammonia molecules.

D The O–H bond in water is stronger than the N–H bond in ammonia.

4 *Use of the Data Booklet is relevant to this question.*

The ^{68}Ge isotope of the Group IV element germanium is medically useful because it undergoes a natural radioactive process to give an isotope, X, which can be used to detect tumours. This transformation of germanium occurs when an electron enters the nucleus, changing a proton to a neutron.

Which statement about the composition of isotope X is correct?

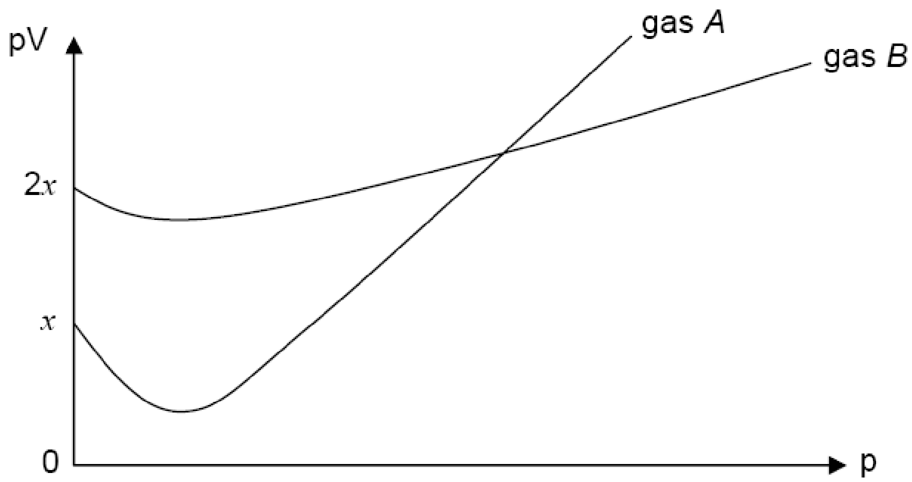
A It has 4 electrons in its outermost shell.

B It has a total of 32 electrons.

C Its proton number is 32.

D It has 37 neutrons.

5 The value of pV is plotted against p for 2 gases, **A** and **B**, where p is the pressure and V is the volume of the gas.



Which of the following could be identities of the gases?

	gas **A**	gas **B**
A	0.5 mol of CH_4 at 25 °C	0.5 mol of CH_4 at 50 °C
B	0.5 mol of NH_3 at 25 °C	0.5 mol of NH_3 at 323 °C
C	0.5 mol of CH_4 at 25 °C	1 mol of NH_3 at 25 °C
D	1 mol of NH_3 at 25 °C	0.5 mol of CH_4 at 25 °C

6 The standard enthalpy change of reaction of $2NO_2(g) \rightarrow N_2O_4(g)$ is x kJ mol^{-1}. The standard enthalpy change of formation of $N_2O_4(g)$ is y kJ mol^{-1}.

What is the standard enthalpy change of formation of $NO_2(g)$?

A $y - x$

B $x - y$

C $\frac{1}{2}y - \frac{1}{2}x$

D $\frac{1}{2}x - \frac{1}{2}y$

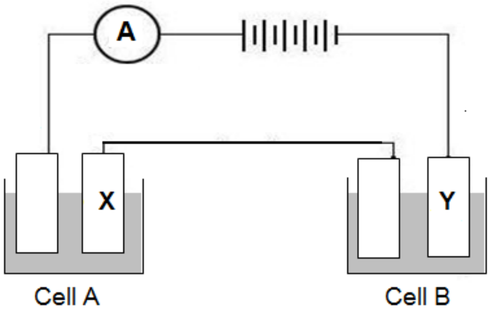
7 The primary structure of haemoglobin is formed from their constituent amino acids. In the presence of the haem groups, the chains spontaneously coalesce into a haemoglobin molecule.

What would be the signs of ΔG, ΔH and ΔS for the overall process?

	ΔG	ΔH	ΔS
A	-	-	-
B	-	+	-
C	+	+	-
D	+	+	+

8 *Use of the Data Booklet is relevant to this question.*

A student carried out an experiment involving the electrolysis of aqueous copper(II) sulfate in cell **A** and aqueous chromium(III) sulfate in cell **B**.



Given that 6.35 g of copper was deposited at electrode **X** at the end of the experiment, what is the mass of chromium deposited at electrode **Y**?

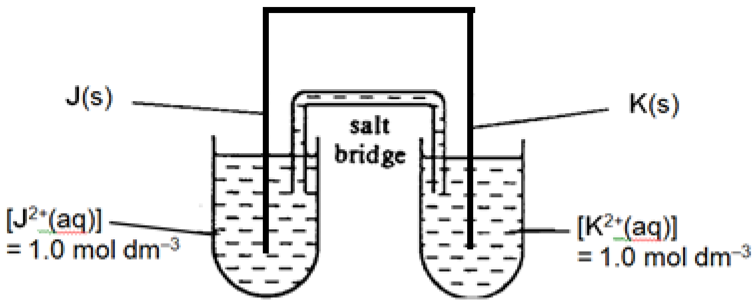
A 0.87 g **B** 1.74 g

C 3.47 g **D** 10.4 g

9 The standard electrode potentials for metals **J** and **K** are given below.

$$\mathbf{J}^{2+}(aq) \mid \mathbf{J}(s) \quad E^\ominus = +0.80\text{ V}$$
$$\mathbf{K}^{2+}(aq) \mid \mathbf{K}(s) \quad E^\ominus = -0.44\text{ V}$$

The electrochemical cell shown in the diagram below is set up.



Which of the following statements about the cell is correct?

A The anions from the salt bridge will enter $\mathbf{K}^{2+}$(aq) | **K**(s) half-cell.

B The electrons flow from electrode **J** to **K**.

C The change in mass for electrode **J** and **K** over a period of time, t, is the same.

D The e.m.f. of the cell will decrease when the concentration of $\mathbf{K}^{2+}$ ions decreases.

10 Ammonium carbamate, $NH_4CO_2NH_2$, decomposes as follows:

$$NH_4CO_2NH_2(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$$

Which of the following statement about this system in equilibrium is correct?

A The value of K_p depends on the amount of $NH_4CO_2NH_2$ present at equilibrium.

B Increasing the volume of the system will shift the equilibrium to the right.

C Decreasing the pressure of the system will increase the equilibrium constant, K_p.

D The use of a catalyst will increase the rate of the reaction by increasing the equilibrium constant, K_p.

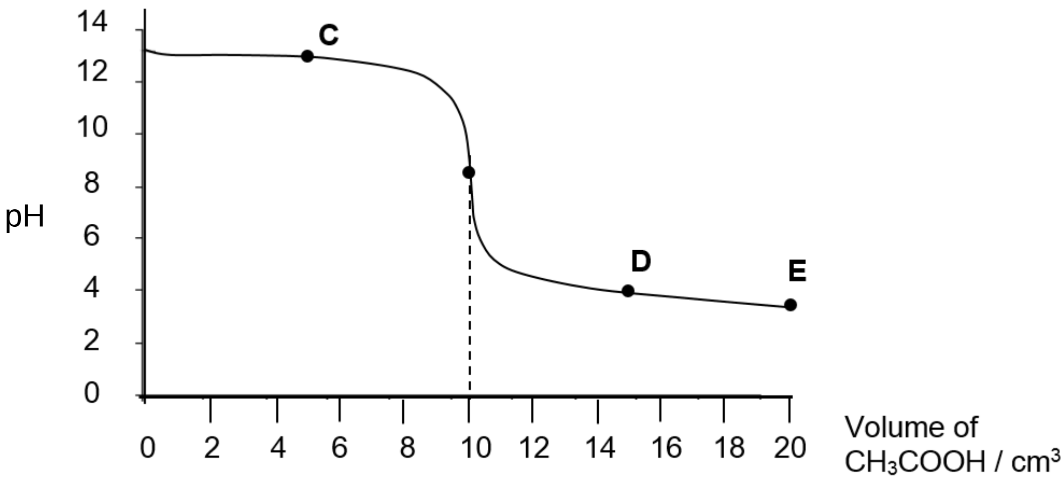
11 Gas **A** decomposes to two other gases, **B** and **C**, according to the equation:

$$2\mathbf{A}(g) \rightleftharpoons 3\mathbf{B}(g) + \mathbf{C}(g) \qquad \Delta H = +x \text{ kJ mol}^{-1}$$

Which of the following correctly describes what will happen if a proposed change is made to this system at equilibrium?

	proposed change	value of K_c	position of equilibrium
A	add inert solid **X**	no change	shift left
B	increase pressure	no change	shift right
C	increase temperature	increase	shift right
D	add catalyst	increase	no change

12 When 5.00 cm^3 of a 0.100 mol dm^{-3} base is titrated against 0.100 mol dm^{-3} of aqueous CH_3CO_2H, the following titration curve is obtained.



Which of the following correctly identifies the base used and the point on the curve at which pH = pK_a?

	base used	point where pH = pK_a
A	KOH(aq)	**C**
B	$Ba(OH)_2$(aq)	**D**
C	KOH(aq)	**E**
D	$Ba(OH)_2$(aq)	**E**

13 The numerical values of the solubility product of calcium carbonate and calcium fluoride at 25 °C are 8.7×10^{-9} and 4.0×10^{-11} respectively.

Which of the following statements is correct?

A Calcium carbonate has a higher solubility than calcium fluoride.

B Addition of sodium fluoride to a saturated solution of calcium fluoride increases the ionic product of calcium fluoride.

C Addition of sodium carbonate to a solution containing calcium carbonate decreases the solubility product of calcium carbonate.

D Addition of calcium nitrate to a solution containing 1 mol dm^{-3} each of carbonate and fluoride ions causes calcium carbonate to precipitate out first.

14 The following experimental results are obtained for a reaction with the general rate equation:

$$\text{rate} = k\,[\mathbf{M}]^x\,[\mathbf{N}]^y\,[\mathbf{L}]$$

experiment	**[M] / mol dm^{-3}**	**[N] / mol dm^{-3}**	**[L] / mol dm^{-3}**	**relative rate**
1	0.1	0.2	0.3	1
2	0.2	0.6	1.2	72

Which of the following are possible values of ***x*** and ***y***?

	x	***y***
A	1	1
B	2	1
C	1	2
D	2	2

15 Element **G** is in the third period of the Periodic Table. The chloride of **G** has a simple molecular structure while the oxide of **G** has a giant ionic structure.

Which of the following statements about **G** is correct?

A The oxide of **G** dissolves in water to give an acidic solution.

B The oxide of **G** is insoluble in aqueous sodium hydroxide.

C The chloride of **G** is used as a refractory material.

D In the gaseous phase, the chloride of **G** forms a dimer.

16 What changes occur in the magnitudes of:

(i) the lattice energy,
(ii) the standard enthalpy change of hydration,
(iii) the solubility of the sulfates,

as Group II is descended?

	lattice energy	$\Delta H^{\ominus}_{hyd}$	solubility of sulfate
A	decreases	decreases	decreases
B	decreases	increases	decreases
C	increases	decreases	increases
D	increases	increases	increases

17 Group VII elements and their hydrogen compounds exhibit trends in properties.

Which of these properties increases down the group $Cl \rightarrow Br \rightarrow I$?

A the volatility of the elements

B the reducing power of hydrogen halides

C the thermal stability of hydrogen halides

D the hydrogen-halogen bond dissociation energy

18 Which of the following reactions of the first row transition metal ions is correct?

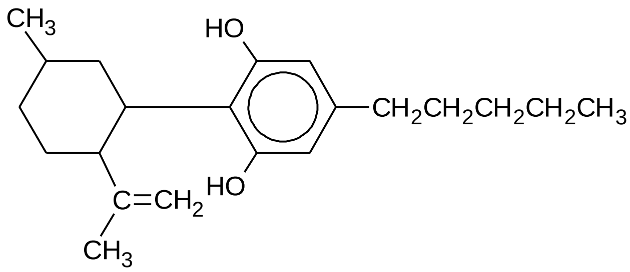
A Addition of NaOH(aq) to a yellow solution of K_2CrO_4 produces an orange solution of $K_2Cr_2O_7$(aq).

B Addition of SCN^-(aq) to Fe^{2+}(aq) produces a blood red solution of $[Fe(H_2O)_5SCN]^+$(aq).

C Addition of Na_2CO_3(aq) to $CrCl_3$(aq) produces a green precipitate of $Cr_2(CO_3)_3$.

D Addition of water to a yellow solution of $[CuCl_4]^{2-}$(aq) produces a light blue solution containing $[Cu(H_2O)_6]^{2+}$(aq).

19 The psychoactive drug in cannabis has the following structure:



How many isomers are there in the final product formed, by reacting the compound above with an excess of Br_2 in an inert solvent, in the absence of UV light?

A 4

B 8

C 16

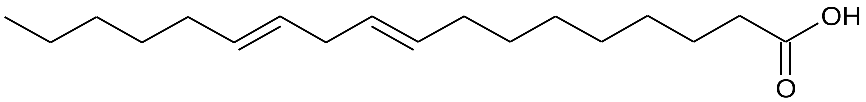
D 32

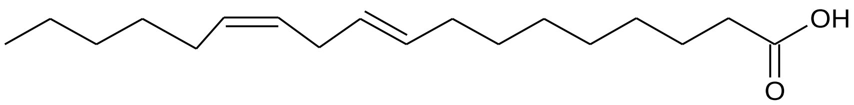
20 Low fat sunflower spreads which are high in polyunsaturated oil, contain esters of linoleic acid.

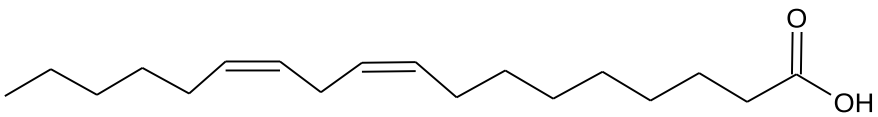
$$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7CO_2H$$

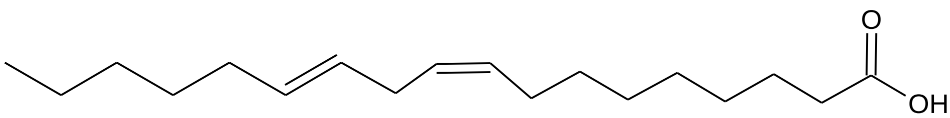
On the lid of a brand of spread it is claimed that the spread contains virtually no *trans* fatty acids.

Which isomer does **not** contain a *trans* linkage and could be present in the spread?

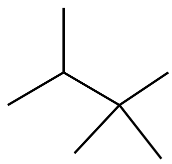
A 

B 

C 

D 

21 Hydrocarbon **X** undergoes free radical substitution with bromine in the presence of ultraviolet light. In a propagation step, the free radical **Y•** is formed by the loss of one hydrogen atom.



+ Br• ⟶ **Y**• + HBr

hydrocarbon **X**

How many possible monobromo-substituted radicals **Y•** can be formed from hydrocarbon **X**?

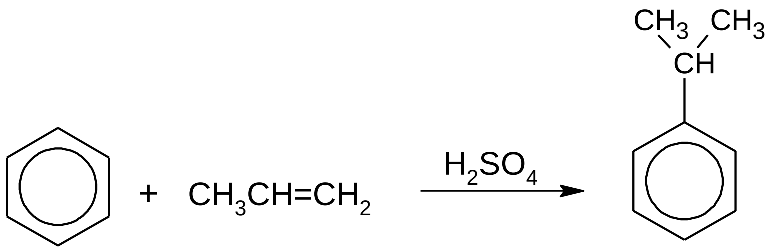
A 2

B 3

C 4

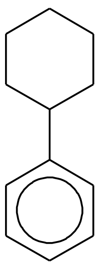
D 5

22 The first stage of the cumene process for the industrial production of phenol is as follows.

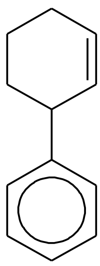


Which of the following would be the product of the reaction, under similar conditions, between benzene and cyclohexene?

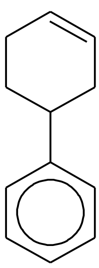
A



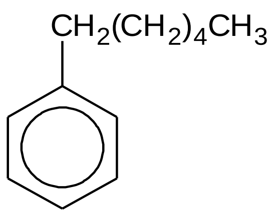
B



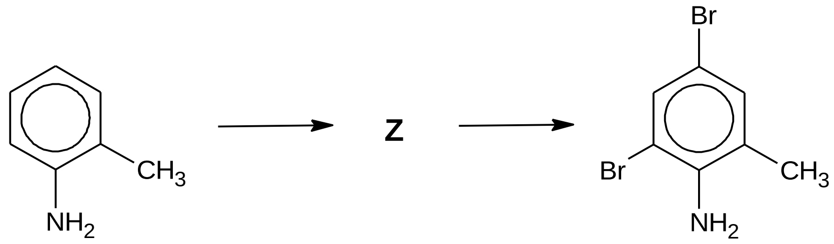
C



D

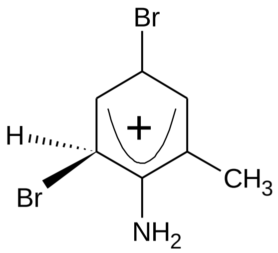


23 When 2-methylphenylamine reacts with an excess of $Br_2(aq)$ to form 4,6-dibromo-2-methylphenylamine, one of the intermediates formed is cation **Z**.

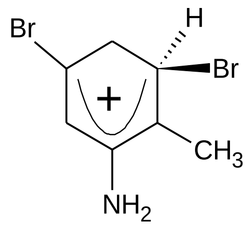


Which of the following could be cation **Z**?

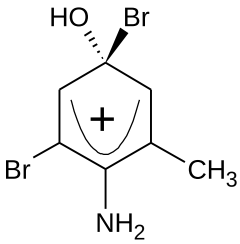
A



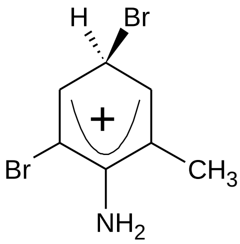
B



C

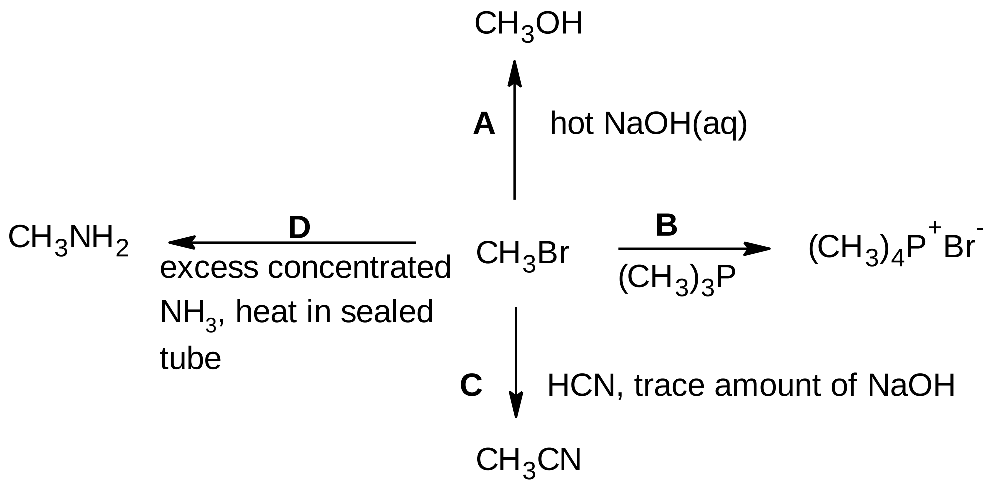


D

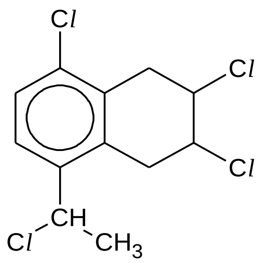


24 A reaction scheme involving bromomethane is given below.

Which of the reactions does **not** take place?

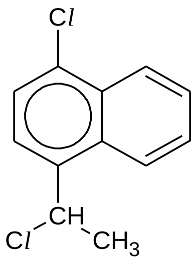


25 Which of the following represents the structure of the organic product when compound **S** is heated with ethanolic sodium hydroxide?

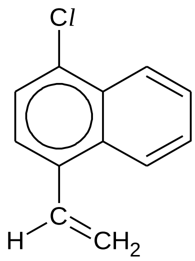


Compound **S**

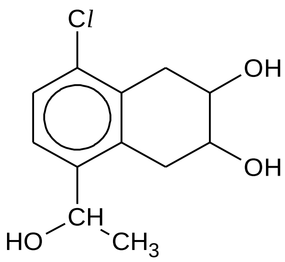
A



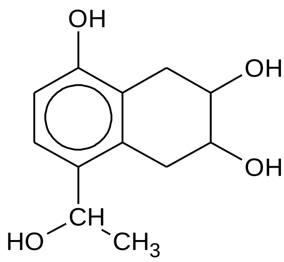
B



C



D



26 Compound **P**, $C_3H_8O_2$, is an alcohol containing two –OH groups per molecule. It can be oxidised in stages to compound **Q**, $C_3H_6O_2$, which reacts with 2,4-dinitrophenylhydrazine but **not** with Fehling's solution. Compound **Q** is then further oxidised to **R**, $C_3H_4O_3$ which reacts with aqueous sodium carbonate.

Which of the following about the properties of **P**, **Q** and **R** is correct?

	has chiral centre	reacts with alkaline aqueous iodine to give tri-iodomethane
A	**P**, **Q** and **R**	**P** only
B	**P** only	**P**, **Q** and **R**
C	**Q** only	**P** and **Q**
D	**P** and **Q**	**Q** and **R**

27 Which reagent reacts with ethanal to give an organic product **without** a π bonded carbon atom?

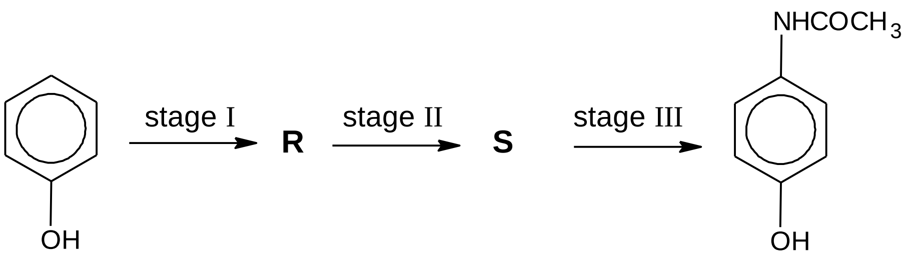
A hydrogen cyanide

B ammoniacal silver nitrate

C potassium dichromate(VI)

D sodium boron hydride

28 Paracetamol is a pain-killing drug.



Paracetamol

What are the reagents and conditions required in this three-stage synthesis?

	stage I	stage II	stage III
A	excess NH_3 in ethanol, heat in sealed tube	(1) Sn, concentrated HC*l* heat (2) NaOH(aq)	CH_3COCl
B	excess NH_3 in ethanol, heat in sealed tube	H_2, nickel catalyst, 200 °C	CH_3CO_2H
C	aqueous HNO_3	(1) Sn, concentrated HC*l* heat (2) NaOH(aq)	CH_3COCl
D	concentrated HNO_3, concentrated H_2SO_4 heat	H_2, nickel catalyst, 200 °C	CH_3CO_2H

29 An ester **P** was heated under reflux with aqueous sodium hydroxide and the resulting mixture was distilled. The distilate gave a positive tri-iodomethane test. The residue in the distillation flask, after acidification, gave a white precipitate.

What could be **P**?

A $CH_3CO_2C_6H_5$

B $C_6H_5CO_2CH_3$

C $C_6H_5CO_2C(CH_3)_3$

D $C_6H_5CO_2CH_2CH_3$

30 Which structure will be present when the amino acid aspartic acid,

$$\begin{array}{ll} H_2N & CHCO_2H \\ & | \\ & CH_2CO_2H \end{array}$$

is in aqueous solution at pH 10?

A
$$\begin{array}{lll} H_3\overset{+}{N}- & CH & -CO_2H \\ & | & \\ & CH_2CO_2H & \end{array}$$

B
$$\begin{array}{lll} H_3\overset{+}{N}- & CH & -CO_2^{\,-} \\ & | & \\ & CH_2CO_2H & \end{array}$$

C
$$\begin{array}{lll} H_3\overset{+}{N}- & CH & -CO_2H \\ & | & \\ & CH_2CO_2^{\,-} & \end{array}$$

D
$$\begin{array}{lll} H_2N- & CH & -CO_2^{\,-} \\ & | & \\ & CH_2CO_2^{\,-} & \end{array}$$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

31 Sodium hydrogensulfide, NaSH, is used to remove hair from animal hides.

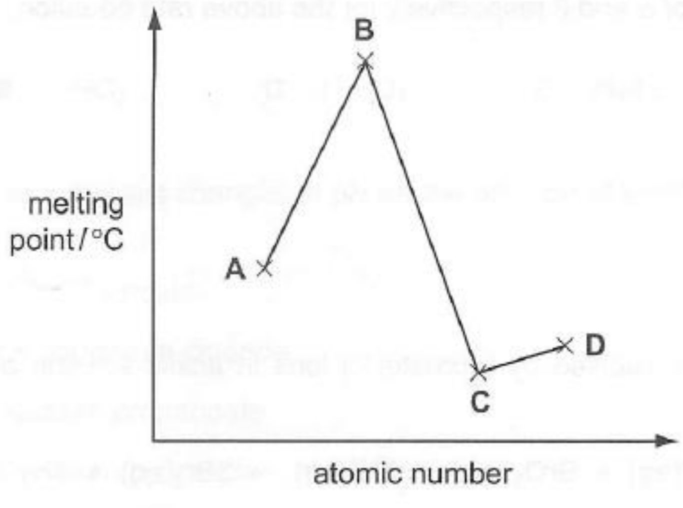
Which statements about the SH^- ion are correct?

1 There are three lone pairs of electrons around the sulfur atom.

2 Its conjugate acid contains a total of 18 electrons.

3 Sulfur has an oxidation state of +2.

32 The graph shows the melting points of four consecutive elements in Period 3, Na to Ar, of the Periodic Table.



Which of the following statements are correct?

1 **A** has the highest electrical conductivity among the four elements.

2 **C** has the highest first ionisation energy among the four elements.

3 **D** burns with a blue flame.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

33 Which statements are likely to be true for astatine (At) which is below iodine in Group VII of the Periodic Table?

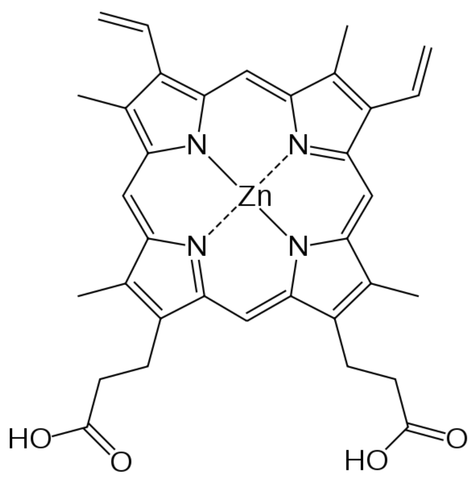
1 At reacts with H_2 vigorously to form HAt.

2 Aqueous NaAt reacts with aqueous $AgNO_3$ to form a precipitate of AgAt which is insoluble in concentrated aqueous NH_3.

3 Solid NaAt reacts with concentrated H_2SO_4 to form At_2.

34 Zinc protoporphyrin (ZPP) is a compound found in red blood cells when haem production is inhibited by the presence of lead or by a lack of iron in blood.

A simplified structure of a molecule of ZPP is shown below.



The zinc atom is situated in the centre of a planar arrangement of four nitrogen atoms.

What does this structure suggest about the bonding around zinc?

1 dative covalent bonding

2 σ bonding

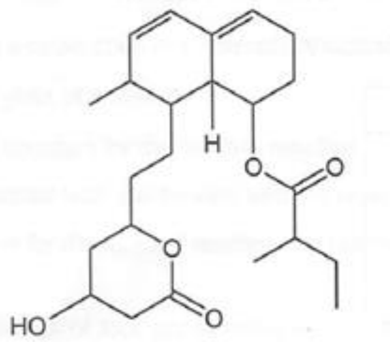
3 sp^3 hybridisation

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

35 *Mevastatin* belongs to the statin group of drugs that are used to lower cholesterol levels in humans.



mevastatin

Which statements about *mevastatin* are correct?

1 It contains seven chiral centres.

2 It contains four sp^2 hybridised carbon atoms.

3 It is soluble in water.

36 Which of the following species are formed in the steps indicated when ethane reacts with chlorine gas in ultraviolet light?

	steps	**species**
1	propagation	$\bullet CH_2CHCl_2$
2	propagation	CH_3CHCl_2
3	termination	$CH_3CH_2CH_2CH_2Cl$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

37 A catalytic converter is part of the exhaust system of many modern cars.

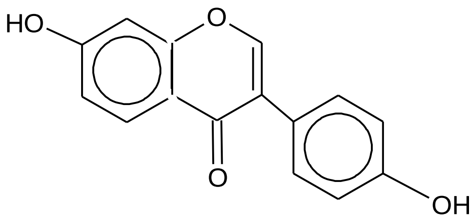
Which reactions does **not** occur in a catalytic converter?

1 $CO_2 + NO \rightarrow CO + NO_2$

2 $2SO_2 + 2NO \rightarrow 2SO_3 + N_2$

3 $2CO + 2NO \rightarrow 2CO_2 + N_2$

38 Soy beans are a good source of dietary isoflavenoids, which are claimed to help in the prevention of some cancers. The major isoflavenoid in soy is diadzein.



diadzein

When diadzein is treated with hydrogen in the presence of a platinum catalyst and the product is then oxidised by warming with acidified $K_2Cr_2O_7$, compound **Q** is formed.

Which statements about compound **Q** are correct?

1 1 mole of compound **Q** reacts with 2 moles of aqueous sodium hydroxide.

2 1 mole of compound **Q** reacts with 4 moles of aqueous bromine.

3 1 mole of compound **Q** reacts with 2 moles of sodium to form 2 moles of hydrogen.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

39 Chlorofluorocarbons (CFCs) have been widely used in aerosol sprays, refrigerators and in making foamed plastics, but are now known to destroy ozone in the upper atmosphere.

What will not destroy ozone, and therefore can be used safely as a replacement for CFCs?

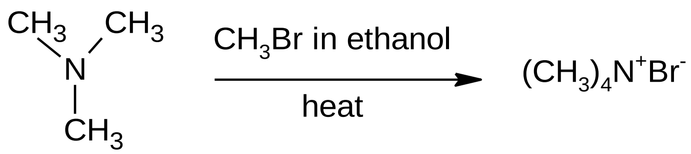
1 $CH_3CH_2CH_2CH_3$

2 CCl_3CBr_3

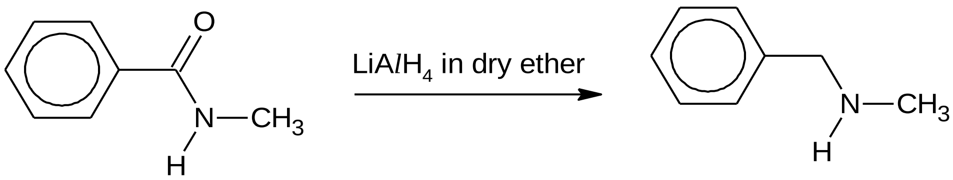
3 $CHClFCClF_2$

40 Which reactions would give the products stated?

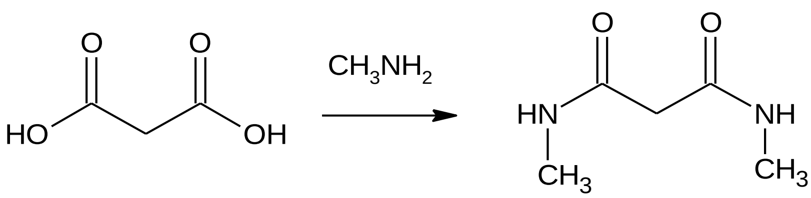
1



2



3



End of paper

Answers

1	B	11	C	21	B	31	B
2	D	12	D	22	A	32	A
3	C	13	B	23	D	33	C
4	D	14	C	24	C	34	B
5	B	15	D	25	B	35	D
6	C	16	A	26	B	36	A
7	A	17	B	27	D	37	B
8	C	18	D	28	C	38	B
9	A	19	C	29	D	39	D
10	B	20	C	30	D	40	B

